THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 8: **Introduction to Trigonometry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $\tan A = \frac{4}{3}$, then $\sin A =$ 1

(a) $\frac{4}{5}$ **(b)** $\frac{4}{3}$ **(c)** $\frac{3}{5}$ **(d)** $\frac{5}{4}$

2. If $\cos A = \frac{5}{13}$, then $\tan A =$ 1

(a) $\frac{12}{13}$ **(b)** $\frac{5}{12}$ **(c)** $\frac{12}{5}$ **(d)** $\frac{13}{5}$

3. The value of $\sin^2 30^\circ + \cos^2 30^\circ$ is: 1

(a) 2 **(b)** 1 **(c)** $\frac{3}{2}$ **(d)** $\frac{1}{4}$

4. $\tan 20^\circ \cdot \tan 70^\circ =$ 1

(a) 0 **(b)** 2 **(c)** 20 **(d)** 1

5. The value of $\sin 15^\circ / \cos 75^\circ$ is: 1

(a) 1 **(b)** 15 **(c)** 2 **(d)** 0

6. $(2\sin\theta)^2 + (2\cos\theta)^2 =$ 1

(a) 5 **(b)** 2 **(c)** 1 **(d)** 4

7. If $\sec\theta = \frac{5}{3}$ then $\cos\theta =$ 1

(a) $\frac{3}{5}$ **(b)** $\frac{5}{3}$ **(c)** $\frac{4}{3}$ **(d)** $\frac{4}{5}$

8. If $\tan A = \frac{12}{5}$, then $\sin A =$ 1

(a) $\frac{5}{13}$ **(b)** $\frac{13}{12}$ **(c)** $\frac{12}{5}$ **(d)** $\frac{12}{13}$

9. If $\cos A = \frac{8}{17}$, then $\tan A =$ 1

(a) $\frac{8}{15}$ **(b)** $\frac{15}{8}$ **(c)** $\frac{17}{8}$ **(d)** $\frac{15}{17}$

10. The value of $\tan^2 60^\circ - \tan^2 30^\circ$ is: 1

(a) $\frac{16}{3}$ **(b)** $\frac{11}{3}$ **(c)** $\frac{8}{3}$ **(d)** 3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If $\tan A = \frac{4}{3}$, find $\sec A$. 2

12. Evaluate: $\dfrac{\sin 25^\circ}{\cos 65^\circ} + \dfrac{\tan 40^\circ}{\cot 50^\circ}$. 2

13. Evaluate: $\dfrac{\tan 21^\circ}{\cot 69^\circ}$. 2

14. If $\tan A = \frac{12}{5}$, find $\sec A$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $\tan\theta = \dfrac{12}{5}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

16. In $\triangle ABC$, right-angled at B, $AB = 7$ cm and $BC = 24$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

17. Prove that: $(1 - \cos^2\theta)\csc^2\theta = 1$. 3

OR

If $\tan\theta = \dfrac{15}{8}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 3

18. In $\triangle ABC$, right-angled at B, $AB = 20$ cm and $BC = 21$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

OR

Prove that: $\tan\theta + \cot\theta = \sec\theta\csc\theta$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that: $\dfrac{\sin\theta}{1+\cos\theta} + \dfrac{1+\cos\theta}{\sin\theta} = 2\csc\theta$. 5

OR

If $\tan A = \dfrac{4}{3}$, evaluate $\dfrac{4\sin A + 3\cos A}{4\sin A + 3\cos A}$. 5

20. Prove that: $(\csc\theta - \cot\theta)^2 = \dfrac{1-\cos\theta}{1+\cos\theta}$. 5

OR

If $\tan A = \dfrac{12}{5}$, evaluate $\dfrac{5\sin A + 2\cos A}{3\sin A + 2\cos A}$. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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